

Supplementary Information

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Supplementary Note S1 | Why FWHM width W is interpreted as an operational uncertainty proxy

In this study, the full width at half maximum (FWHM), denoted as W , was used as an operational measure of observable width in biological-process transcriptomic observables. This section clarifies why W was interpreted as an uncertainty-related quantity, what this interpretation means, and what it does not mean.

The D-PHY framework does not assume that transcriptomic measurements directly reveal the complete biological state of a tumor. Instead, a measured transcriptomic observable is treated as a projected representation of an underlying latent biological system. In simplified form, this can be written as:

$$O = h(S) + \epsilon$$

where S represents the latent biological organization, $h(S)$ represents the observation process, O represents the measured observable, and ϵ represents unresolved variability and measurement uncertainty.

Under this view, a biological-process observable is not a direct copy of the underlying biological state. It is an effective projection shaped by tumor heterogeneity, cellular mixture, pathway overlap, regulatory coupling, and measurement limitations.

For a biological process B , the corresponding pathway-level observable was represented by two quantities:

$$(\mu_B, W_B)$$

where μ_B represents the observable center or effective activity magnitude, and W_B represents the distributional width of the observable. In this study, W_B was estimated using FWHM.

The reason for using FWHM is that it provides a simple and interpretable measure of how broadly a biological-process observable is distributed. A narrow distribution indicates that the observable is concentrated around a relatively well-defined activity region, whereas a broad distribution indicates that the observable is dispersed across a wider range of values.

In the D-PHY interpretation, this width is treated as an operational proxy for uncertainty or “state-cloud thickness.”

This interpretation is operational rather than literal. Specifically, W does not represent thermodynamic entropy, exact information-theoretic entropy, or a direct measurement of the number of hidden biological states. Instead, W reflects the observed spread of a projected biological-process signal. This spread may arise from several sources, including:

- tumor cellular heterogeneity;
- variation in tumor purity and microenvironmental composition;
- pathway-level gene-expression dispersion;
- pathway overlap and regulatory cross-talk;
- unresolved latent-state differences;
- measurement and sampling variability.

Therefore, W should be interpreted as an observable-level uncertainty proxy, not as a direct physical entropy.

This distinction is important. A high W value does not prove that the underlying biological system has high entropy in a strict thermodynamic sense. Rather, it indicates that the observed pathway-level projection is broad or diffuse. In a partially observable system, such broadness can be interpreted as increased ambiguity in how the observable maps back to possible latent biological configurations.

Conversely, a low W value does not prove that the latent biological system is

simple or homogeneous. It only indicates that, under the current transcriptomic projection and pathway definition, the observable appears more concentrated.

Thus, the interpretation is deliberately conservative:

$$W \approx \text{observable projection width}$$

and operationally:

$$W \sim \text{state-cloud thickness proxy}$$

This is why W was combined with activity magnitude μ to define observable coherence:

$$C_B = \frac{|\mu_B|}{W_B + \delta}$$

where δ is a small stabilization constant. This formulation captures the idea that an observable with strong activity but very broad uncertainty width may be less informative than an observable with comparable activity but narrower uncertainty structure.

In summary, FWHM-based width W was used because it provides a practical and reproducible way to quantify the spread of pathway-level transcriptomic observables. Within the D-PHY framework, this spread is interpreted as an operational uncertainty proxy arising from partial observability, not as a literal entropy or complete latent-state measure.

This interpretation allows transcriptomic pathway observables to be analyzed not only by their activity magnitude, but also by the uncertainty structure associated with their projected representation.

Supplementary Note S2 | Why survival was used as an operational endpoint under partial observability

One of the central challenges in cancer-system analysis is that the true latent

biological state of a tumor is not directly observable. Ideally, a framework intended to study state evolution or progression would analyze explicit latent-state trajectories:

$$S(t) \rightarrow S(t + 1) \rightarrow S(t + 2)$$

where $S(t)$ represents the evolving biological organization of the tumor system over time.

However, in large public cancer cohorts such as TCGA, true longitudinal latent-state measurements are generally unavailable. Most datasets consist of static bulk transcriptomic observations collected at a single clinical time point, together with downstream clinical outcomes such as survival or recurrence.

Consequently, the D-PHY framework does not directly reconstruct latent-state dynamics. Instead, it evaluates whether transcriptomic observables preserve clinically relevant separation structure under partial observability.

Within this context, clinical survival was used as an operational endpoint.

Importantly, survival was not interpreted as a direct measurement of latent biological state. Instead, survival functions as a downstream clinical consequence that may indirectly reflect differences in underlying biological organization.

The central logic is therefore operational rather than deterministic:

Observable differences $\rightarrow$ Outcome-associated separation
rather than:

Observable differences $\rightarrow$ Direct latent-state reconstruction

This distinction is critical.

The D-PHY framework does not assume that transcriptomic pathway observables uniquely determine tumor state or future trajectory. Rather, the framework

evaluates whether certain observable organizations are associated with more separable clinical outcomes than others.

For this reason, clinical discriminability was defined operationally as:

$$D_{\text{clinical}} = -\log_{10}(p_{\text{log-rank}})$$

where the log-rank p-value was obtained from Kaplan–Meier survival analysis after patient stratification.

Under this definition, higher discriminability indicates stronger outcome-associated separation under the current observational framework. It does not imply that the underlying latent biological system has been uniquely identified.

The use of survival as an operational endpoint was motivated by several practical considerations.

First, survival is one of the most consistently available clinical endpoints across large TCGA cohorts. This allows systematic cross-cancer comparison under a shared analysis framework.

Second, survival integrates multiple downstream biological consequences, including:

- tumor aggressiveness;
- therapy response;
- resistance emergence;
- metastatic progression;
- microenvironmental interactions.

Although survival is therefore coarse and indirect, it still contains clinically meaningful information about long-term system behavior.

Third, survival-based evaluation provides a practical way to assess whether a transcriptomic observable preserves clinically relevant structure. In other words,

survival acts as an external validation layer for observable usefulness.

Within the D-PHY framework, the key question is therefore not:

“Can transcriptomics uniquely reconstruct the hidden tumor state?”

but rather:

“Does a given observable organization preserve clinically meaningful separation under partial observability?”

This distinction is fundamental.

Because transcriptomic observations are projections:

$$O = h(S) + \epsilon$$

multiple latent biological configurations may produce similar observable measurements. Under such conditions, direct latent-state inference becomes intrinsically underdetermined.

Survival analysis therefore provides an operational strategy for evaluating observable quality without requiring explicit reconstruction of the hidden biological state.

However, several important limitations must be emphasized.

First, survival is not equivalent to latent-state evolution. Two patients with similar survival outcomes may still possess substantially different biological organizations.

Second, survival is influenced by many non-transcriptomic factors, including:

- treatment differences;
- clinical management;
- demographic variability;
- comorbidities;
- sampling variability.

Third, survival is a relatively coarse endpoint. It compresses complex biological trajectories into a limited outcome measure.

Accordingly, the D-PHY framework treats survival as an operational proxy rather than a complete representation of tumor-system dynamics.

This limitation is explicitly acknowledged throughout the study.

The present framework therefore represents an observable-centered analysis strategy rather than a direct dynamical reconstruction framework.

Future extensions of the D-PHY framework may incorporate more direct trajectory-related datasets, including:

- longitudinal transcriptomics;
- treatment time-course datasets;
- spatial transcriptomics;
- single-cell temporal measurements;
- therapy-response trajectories.

Such datasets may allow future uncertainty-aware frameworks to evaluate:

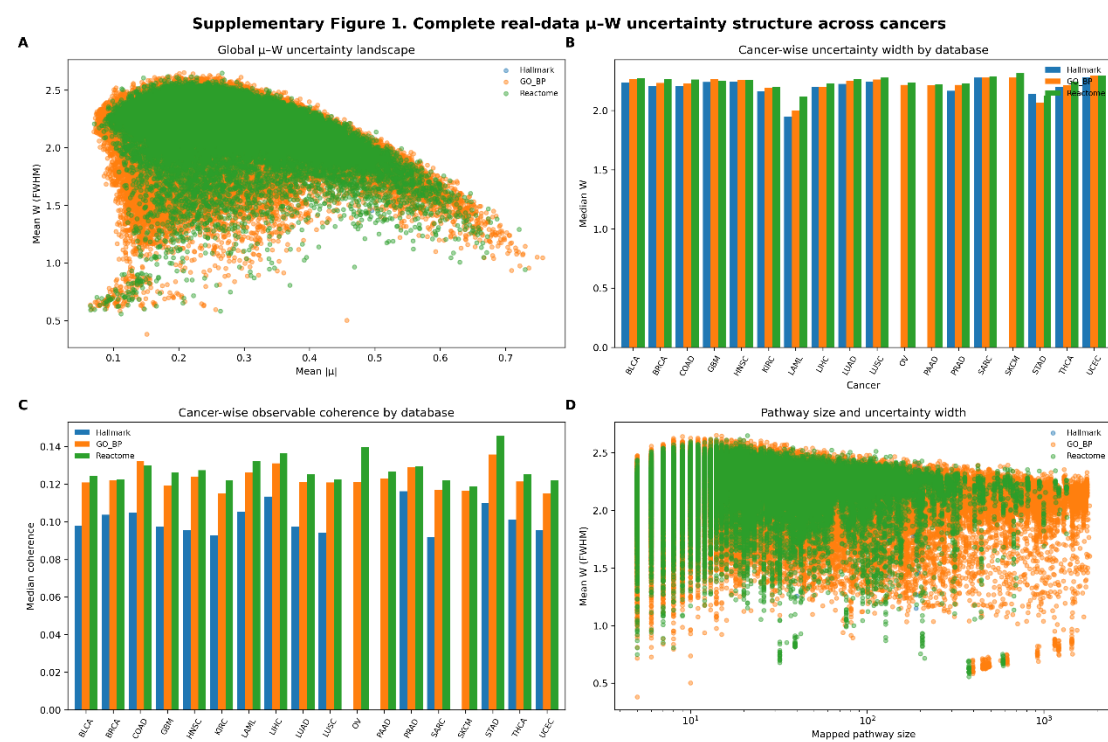
$$S(t) \rightarrow S(t + 1)$$

more directly.

Nevertheless, under the current availability of large-scale public cancer datasets, survival remains one of the most practical and reproducible operational endpoints for evaluating whether transcriptomic observables preserve clinically meaningful structure under partial observability.

Within this context, the use of survival in the present study should therefore be interpreted as a pragmatic observable-validation strategy rather than a claim of complete latent-state inference.

Supplementary Note S3 | Why pathway-level averaging remains biologically useful under partial observability



Supplementary Figure S1 | Complete real-data μ -W uncertainty structure across cancers.

(A) Global μ -W uncertainty landscape constructed from Hallmark, GO:BP, and Reactome gene-set observables across all analyzed cancer cohorts. Each point represents one observable gene set, where μ denotes the mean absolute activity magnitude and W denotes the corresponding uncertainty width estimated using the full width at half maximum (FWHM) of the observable distribution. The resulting landscape reveals a nontrivial organization between observable activity and uncertainty width, indicating that strong observable magnitude does not necessarily imply low uncertainty.

(B) Cancer-wise median uncertainty width across different biological databases. Median W values remain relatively stable across cancers and databases, although modest cancer-specific variations are observed.

(C) Cancer-wise observable coherence estimated as $|\mu|/(W+\delta)$. Coherence distributions differ across cancer types and database structures, suggesting that

pathway organization influences effective observable compactness.

(D) Relationship between mapped pathway size and uncertainty width. Larger pathway sets generally exhibit broader uncertainty distributions and reduced coherence, indicating that observable scale influences the effective uncertainty structure of gene-set representations.

Together, these analyses support the interpretation of pathway-level observables as structured uncertainty clouds rather than deterministic state descriptors, consistent with the AIDO-D-PHY framework under partial observability.

A common concern in transcriptomic pathway analysis is that pathway-level averaging may oversimplify biological organization. Biological pathways contain heterogeneous genes with different functional roles, regulatory dynamics, interaction structures, and context-dependent activities. Consequently, reducing a pathway to a single summary value may appear biologically reductive.

This concern is valid and represents an important limitation of any pathway-level abstraction framework.

However, despite this limitation, pathway-level averaging remains useful within the context of partially observable biological systems. The objective of this note is to clarify why coarse-grained pathway observables were still employed in the D-PHY framework and how they should be interpreted.

The central point is that the D-PHY framework does not interpret pathway averages as exact mechanistic representations of biological pathways. Instead, pathway-level observables are treated as effective coarse-grained projections of underlying biological organization.

In statistical physics and complex-systems analysis, coarse-graining is commonly used when direct access to microscopic states is unavailable or impractical. Rather than modeling every microscopic interaction explicitly, systems are

represented using reduced observables that capture dominant large-scale structure.

Examples include:

- temperature as an effective average kinetic quantity;
- magnetization as an effective spin-order parameter;
- density as an effective many-particle representation.

Importantly, these coarse-grained variables do not preserve all microscopic information. Instead, they summarize aspects of system organization relevant at a particular observational scale.

The pathway-level observables used in the D-PHY framework play a similar role.

Transcriptomic measurements in bulk tumors are inherently coarse-grained already. A bulk transcriptomic profile reflects the combined contribution of:

- tumor cells;
- stromal cells;
- immune infiltration;
- vascular components;
- clonal heterogeneity;
- unresolved spatial structure.

Consequently, the measured transcriptomic profile is itself an effective projection:

$$O = h(S) + \epsilon$$

where the underlying latent biological organization S is only partially observable.

Under such conditions, pathway-level averaging serves as an additional structured reduction layer intended to improve interpretability and stability.

For a biological process B , the pathway-level observable was defined as:

$$\mu_B = \frac{1}{|B|} \sum_{g \in B} z_g$$

where:

- z_g denotes z-scored gene expression,
- $|B|$ denotes pathway size.

Importantly, this average is not interpreted as a direct measurement of pathway activity in a mechanistic sense. Rather, it represents an effective collective projection of pathway-associated transcriptomic behavior.

The usefulness of such averaging arises from several properties.

First, pathway-level aggregation reduces gene-level stochastic variability. Individual transcriptomic measurements are often noisy, unstable, and highly context dependent. Averaging across biologically related genes may partially suppress idiosyncratic fluctuations while preserving broader process-level structure.

Second, pathway-level observables improve cross-cohort interpretability. Individual-gene effects are often difficult to interpret reproducibly across cancers or datasets, whereas pathway-level organization is generally more biologically stable.

Third, pathway-level observables provide a natural interface for uncertainty analysis. The D-PHY framework does not analyze pathway means alone. Instead, it jointly evaluates:

$$(\mu_B, W_B)$$

where:

- μ_B represents pathway-center activity,
- W_B represents observable width.

Thus, the framework explicitly incorporates pathway-level uncertainty structure

rather than assuming that the pathway average alone fully characterizes biological organization.

This distinction is important.

A pathway with moderate activity but narrow uncertainty width may preserve more coherent observable organization than a pathway with high average activity but broad uncertainty dispersion.

Accordingly, the D-PHY framework interprets pathway-level observables geometrically rather than purely descriptively.

Importantly, the present framework does not claim that pathway averaging preserves complete biological detail. Several limitations remain.

First, pathway genes may participate in multiple biological processes simultaneously. Pathway overlap therefore introduces projection ambiguity.

Second, pathway averaging ignores:

- network topology;
- causal directionality;
- regulatory hierarchy;
- dynamic signaling structure.

Third, transcriptomic averaging compresses spatial and cellular heterogeneity into a reduced observable representation.

Fourth, different genes within the same pathway may exhibit opposite or context-dependent regulatory behavior.

These limitations are real and unavoidable within simplified pathway-level representations.

However, the D-PHY framework does not attempt to eliminate these limitations completely. Instead, it explicitly studies how uncertainty structure emerges under such projected representations.

The use of pathway-level observables should therefore be interpreted pragmatically.

The objective is not to reconstruct complete biological mechanisms from pathway averages. Rather, the objective is to evaluate whether certain observable organizations preserve clinically relevant structure more effectively than others under partial observability.

In this sense, pathway-level averaging acts as a biologically informed coarse-graining procedure.

This interpretation is consistent with the broader philosophy of the D-PHY framework:

- transcriptomic observations are projections;
- pathway observables are uncertainty-bearing effective variables;
- discriminability reflects observable usefulness rather than exact latent-state recovery.

Future extensions may incorporate more detailed biological structure, including:

- network topology;
- protein–protein interaction organization;
- signaling directionality;
- spatial transcriptomics;
- single-cell pathway decomposition.

Such approaches may improve observable fidelity and reduce projection ambiguity.

Nevertheless, under the current availability of large-scale transcriptomic cancer datasets, pathway-level averaging remains a practical, interpretable, and reproducible strategy for constructing biologically meaningful observables suitable for uncertainty-aware analysis under partial observability.

Supplementary Note S4 | Why high-discriminability random observables do not invalidate biological structure

One of the most important concerns in transcriptomic outcome analysis is that random gene sets can sometimes produce significant survival separation. This observation was highlighted prominently by previous studies showing that many randomly generated gene signatures may appear statistically associated with clinical outcomes in cancer datasets [1].

At first glance, this phenomenon may seem to undermine the biological relevance of pathway-based transcriptomic analysis. If random gene sets can also produce high discriminability, then an important question arises:

Does pathway-level discriminability merely reflect statistical artifacts rather than meaningful biological organization?

The D-PHY framework explicitly addresses this issue through the structured-versus-random validation analysis.

Importantly, the existence of high-discriminability random observables was not treated as a failure of the framework. Instead, it was interpreted as evidence that high-dimensional transcriptomic spaces already possess nontrivial statistical and biological structure.

This distinction is central to the interpretation of the random-test results.

The D-PHY framework does not assume that transcriptomic measurements form an unstructured random space. On the contrary, transcriptomic systems are highly organized biological systems shaped by:

- regulatory networks;
- pathway interactions;
- cell-state programs;
- tumor microenvironment structure;
- clonal heterogeneity;
- evolutionary selection.

Consequently, even random subsets of genes sampled from such systems may partially capture latent biological axes.

This is especially likely in high-dimensional transcriptomic datasets where many genes exhibit correlated behavior through shared regulatory or pathway structure.

Under these conditions, a random gene set is not truly “random” in a biological sense. Instead, it is sampled from an already structured transcriptomic manifold.

Within the D-PHY framework, this observation is interpreted geometrically.

Transcriptomic observations are represented as projected observables:

$$O = h(S) + \epsilon$$

where:

- S denotes latent biological organization,
- $h(S)$ denotes the projection process,
- O denotes observed transcriptomic structure.

Because the latent biological system itself contains correlated organization, random projections may still preserve fragments of clinically relevant structure.

Therefore, the existence of high-discriminability random observables does not imply that biological pathways are meaningless. Rather, it indicates that transcriptomic spaces contain intrinsic latent organization capable of partially surviving random projection.

This interpretation differs fundamentally from the idea that all transcriptomic discriminability is purely statistical noise.

To evaluate whether curated biological pathways preserve additional structure beyond random expectation, the D-PHY framework employed structured-versus-random comparisons.

For each pathway:

$$B_i$$

size-matched random gene sets:

$$R_i$$

were generated from the same transcriptomic gene universe.

Critically, random sets and structured pathways were processed using identical workflows:

- identical preprocessing;
- identical pathway scoring;
- identical uncertainty-width estimation;
- identical survival analysis;
- identical discriminability computation.

This symmetry is important because it prevents pipeline asymmetry from artificially favoring structured pathways.

The key question was therefore not whether random gene sets can produce discriminability. Instead, the key question was:

Do biologically structured pathways exhibit different uncertainty organization and discriminability distributions compared with random observables?

The results showed that:

- random observables can occasionally achieve high discriminability;
- however, structured pathways exhibit systematically different global behavior.

In particular, curated pathways more frequently produced:

- coherent uncertainty structure;
- stable discriminability;
- reproducible cross-cancer behavior;
- enriched high-discriminability regions.

Thus, the biological contribution emerges not from eliminating randomness entirely, but from reshaping the observable landscape.

Within the D-PHY interpretation, random observables define a baseline transcriptomic geometry, whereas curated pathways represent biologically constrained reorganization of that geometry.

This distinction can be conceptualized as:

Random geometry → Biologically constrained geometry

rather than:

Noise → Biological signal

This interpretation is important because transcriptomic systems are not simple additive signal-versus-noise systems. Instead, they are structured high-dimensional biological systems in which:

- statistical organization,
- pathway organization,
- regulatory structure,

- latent biological variability
- are deeply intertwined.

The D-PHY framework therefore avoids the overly simplistic interpretation that: “random significance invalidates biology.”

Instead, the framework treats random discriminability as part of the intrinsic geometry of high-dimensional transcriptomic spaces.

The structured-versus-random analysis was designed precisely to quantify how biological organization reshapes this geometry.

Importantly, the framework also distinguishes between:

- isolated high-discriminability random events,
- systematic observable organization.

A single random gene set with high discriminability does not necessarily imply reproducible or biologically stable structure.

Accordingly, the D-PHY framework additionally examined:

- distributional behavior across many random sets;
- random-baseline statistics;
- cross-cancer recurrence patterns;
- coherence structure;
- scale-dependent organization.

These analyses collectively provide a more stable characterization of transcriptomic observable structure than isolated high-D observations alone.

Nevertheless, several limitations remain.

First, random and structured observables are both ultimately derived from the

same transcriptomic data space. Therefore, complete separation between “biological” and “statistical” structure is not possible.

Second, certain curated pathways may themselves contain highly correlated genes, which can naturally enhance discriminability.

Third, transcriptomic datasets contain cohort-specific correlations that may influence both structured and random observables.

For these reasons, the D-PHY framework does not interpret pathway discriminability as definitive mechanistic proof.

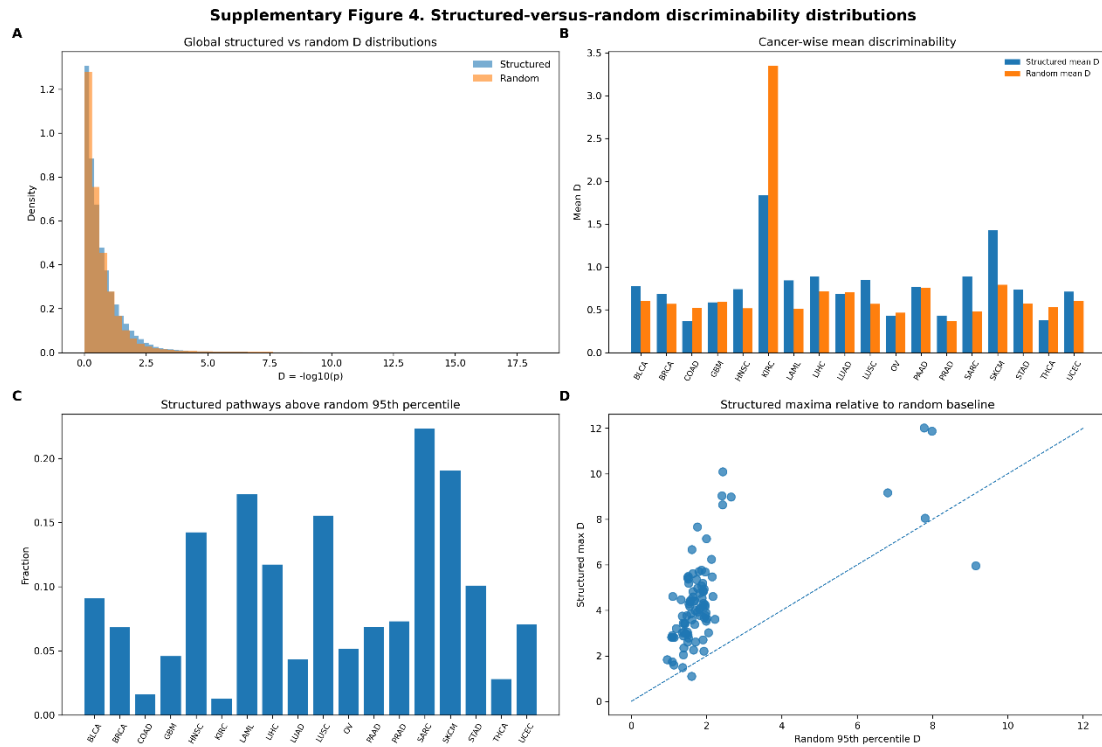
Instead, pathway discriminability is interpreted operationally as evidence that certain observable organizations preserve clinically relevant structure more effectively than others under partial observability.

Within this interpretation, the structured-versus-random framework acts as a calibration layer for evaluating observable organization.

The central conclusion is therefore not that random discriminability invalidates biology, but rather:

biological pathways reshape and constrain the observable geometry of already structured transcriptomic spaces.

Under this view, high-dimensional transcriptomic systems are neither purely random nor perfectly deterministic. Instead, they exhibit partially organized observable geometry shaped jointly by biological structure and statistical projection effects under partial observability.



Supplementary Figure S2 | Structured-versus-random validation of uncertainty-aware discriminability across cancers.

(A) Distribution of discriminability scores obtained from biologically structured pathway observables and size-matched random gene-set observables across multiple cancer cohorts. Although random observables can occasionally produce moderate discriminability, biologically structured pathways exhibit systematically distinct discriminability organization.

(B) Cancer-wise comparison between structured pathway observables and random baselines. Structured observables more frequently preserve reproducible high-discriminability regions across cancers, whereas random observables display broader stochastic variability.

(C) Relationship between uncertainty width W and discriminability for structured and random observables. Structured pathways exhibit more organized uncertainty geometry compared with random gene-set projections.

(D) Representative examples of structured-versus-random observable organization under the D-PHY framework. Curated biological pathways preserve more coherent observable structure and clinically relevant separation than size-matched random projections generated under identical analytical procedures.

Together, these analyses indicate that high-dimensional transcriptomic spaces possess intrinsic structure, but biologically curated pathway organization systematically reshapes the observable geometry and discriminability landscape beyond random expectation.

Reference

1. Most random gene expression signatures are significantly associated with breast cancer outcome

Supplementary Note S5 | Observable geometry, uncertainty compression, and discriminability under partial observability

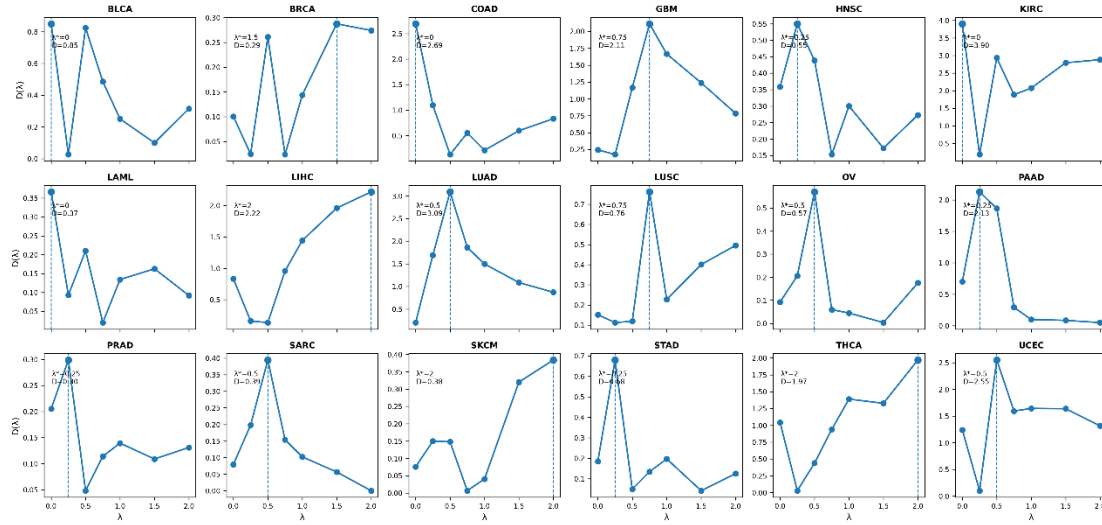
The D-PHY framework was developed to analyze transcriptomic observables under conditions of partial observability. Rather than treating pathway-level transcriptomic measurements as direct representations of latent biological states, the framework interprets them as projected observables with associated uncertainty structure.

This note clarifies the conceptual relationship between:

- observable geometry,
- uncertainty width,
- coherence,
- discriminability,
- uncertainty compression.

The objective is to explain why clinically useful transcriptomic observables are interpreted in the D-PHY framework as geometrically organized uncertainty-bearing projections rather than purely descriptive molecular measurements.

Supplementary Figure 2. Full λ -scan discriminability curves across cancers



Supplementary Figure S3 | Full λ -scan analysis of uncertainty-penalized discriminability across cancers and pathway databases.

(A) Cancer-wise λ -scan curves showing the relationship between uncertainty-penalty strength (λ) and the resulting discriminability score D for pathway-level observables. D was evaluated using survival-based outcome separation under the D_W -PHY framework. Different cancers exhibit distinct λ -dependent response profiles, indicating that optimal uncertainty penalization is cancer- and observable-dependent rather than universal.

(B) Distribution of optimal λ values identified across cancers and biological databases. The optimal λ values span a broad but nonrandom range, suggesting that uncertainty penalization contributes meaningfully to observable selection and discriminability optimization.

(C) Maximum discriminability achieved under λ optimization for each cancer cohort. Several cancers exhibit substantial improvement after uncertainty-aware penalization, whereas others show only modest gains, indicating heterogeneous sensitivity to uncertainty structure.

(D) Relationship between optimal λ and the corresponding maximum discriminability. The absence of a simple monotonic relationship suggests that uncertainty penalization interacts nonlinearly with observable organization and cancer-specific structure.

These results support the interpretation that uncertainty width W contributes nontrivially to discriminability and that incorporating uncertainty-aware

penalization can improve the separation of clinically relevant observable states.

1. Observable geometry under projection

In the D-PHY framework, transcriptomic observations are represented as projections:

$$O = h(S) + \epsilon$$

where:

- S denotes latent biological organization,
- $h(S)$ denotes the observation mapping,
- O denotes the measured observable,
- ϵ denotes unresolved variability and uncertainty.

Because the observation mapping is incomplete and many-to-one, multiple latent biological configurations may produce similar observable measurements. Consequently, transcriptomic observables should not be interpreted as exact latent-state measurements.

Instead, they are interpreted as projected observable regions within an uncertainty-bearing observable space.

For a biological-process observable B , the D-PHY framework represents the observable geometrically using:

$$(\mu_B, W_B)$$

where:

- μ_B denotes the observable center,
- W_B denotes observable width.

This pair defines the effective geometry of the projected pathway observable.

2. Observable center and observable width

The observable center:

$$\mu_B$$

was computed as the mean pathway-level transcriptomic activity.

Operationally, μ_B describes the average location of the observable cloud.

However, pathway activity alone is insufficient for evaluating observable usefulness. Two pathways with similar average activity may exhibit very different uncertainty structures.

To characterize this uncertainty organization, the D-PHY framework additionally estimates observable width:

$$W_B$$

using FWHM-based distributional width analysis.

The width term reflects how broadly the projected observable is distributed.

Operationally:

- narrow width indicates concentrated observable organization,
- broad width indicates diffuse observable organization.

Importantly, W_B is not interpreted as exact entropy or latent-state multiplicity. Rather, it acts as an operational uncertainty-width proxy reflecting the spread of the projected observable.

3. Observable overlap and uncertainty ambiguity

Under partial observability, the central inferential problem is not simply measurement noise. Instead, the primary difficulty arises from observable overlap.

Different latent biological organizations may project into overlapping observable regions.

Geometrically, this means that observable clouds corresponding to different

latent configurations may partially overlap in transcriptomic observable space.

When observable overlap is large:

- latent-state ambiguity increases,
- transcriptomic separability decreases,
- discriminability weakens.

Conversely, when observable organization is more concentrated and overlap is reduced:

- observable separation becomes clearer,
- uncertainty ambiguity decreases,
- discriminability improves.

Thus, discriminability is interpreted geometrically as a property of observable organization under projection.

4. Uncertainty compression

Within the D-PHY framework, coherence is interpreted as a form of uncertainty compression.

Coherence was defined as:

$$C_B = \frac{|\mu_B|}{W_B + \delta}$$

where:

- μ_B denotes observable activity magnitude,
- W_B denotes uncertainty width,
- δ is a small stabilization constant.

This formulation captures the balance between:

- observable magnitude,
- uncertainty dispersion.

High coherence does not necessarily imply that the underlying biological system is simple. Rather, it indicates that the projected observable remains relatively concentrated under the current observation mapping.

Operationally:

High coherence

corresponds to:

- concentrated observable geometry,
- reduced overlap,
- stronger uncertainty compression.

Low coherence

corresponds to:

- diffuse observable geometry,
- increased overlap,
- weaker uncertainty compression.

Thus, coherence measures how effectively a transcriptomic observable preserves organized structure under projection.

5. Relationship between uncertainty compression and discriminability

The D-PHY framework proposes that clinically useful observables emerge when uncertainty structure remains sufficiently compressed under projection.

This idea motivates the uncertainty-aware scoring formulation:

$$D_{W-PHY} = M - \lambda U$$

where:

- M represents observable activity magnitude,
- U represents uncertainty contribution,
- λ controls uncertainty penalization strength.

Under this framework:

- activity alone is insufficient,

- uncertainty organization also matters.

An observable with strong activity but highly diffuse uncertainty structure may exhibit weak clinical separability because observable overlap remains large.

Conversely, an observable with moderate activity but concentrated uncertainty structure may preserve stronger discriminability.

Thus, discriminability emerges not solely from signal magnitude, but from the geometry of projected uncertainty-bearing observables.

6. Observable geometry and scale dependence

The same logic extends naturally to observation scale.

The D-PHY framework defines:

$$K$$

as the number of pathway observables included in the analysis.

Increasing observational scale may simultaneously:

- increase biological coverage,
- increase uncertainty overlap,
- increase projection redundancy.

Consequently:

$$D(K)$$

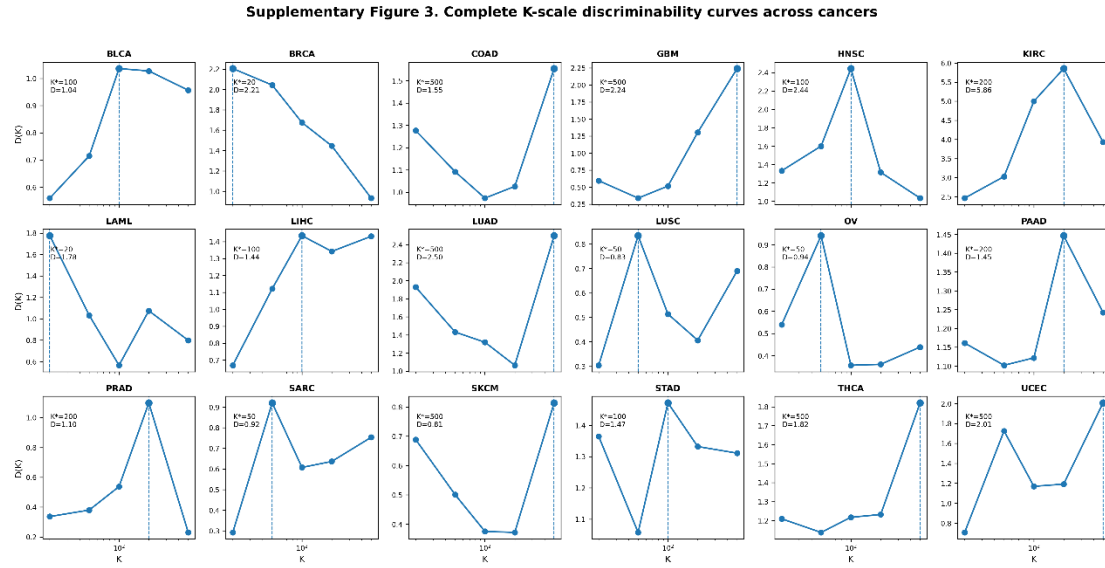
is not expected to increase monotonically.

Instead, optimal observation scales may emerge when the balance between:

- informational coverage,
- uncertainty accumulation

is most favorable.

Under this interpretation, scale dependence is a geometric property of projected observable organization rather than merely a statistical sampling effect.



Supplementary Figure S4 | Full K-scale analysis of observation-scale-dependent discriminability across cancers and pathway databases.

(A) Cancer-wise K-scale curves showing the relationship between observation scale K and discriminability under the D-PHY framework. Here, K denotes the number of pathway-level observables included in the uncertainty-aware observable representation. Distinct cancers exhibit different scale-dependent discriminability profiles, indicating that clinically relevant observable organization depends on observation scale rather than increasing monotonically with additional pathway inclusion.

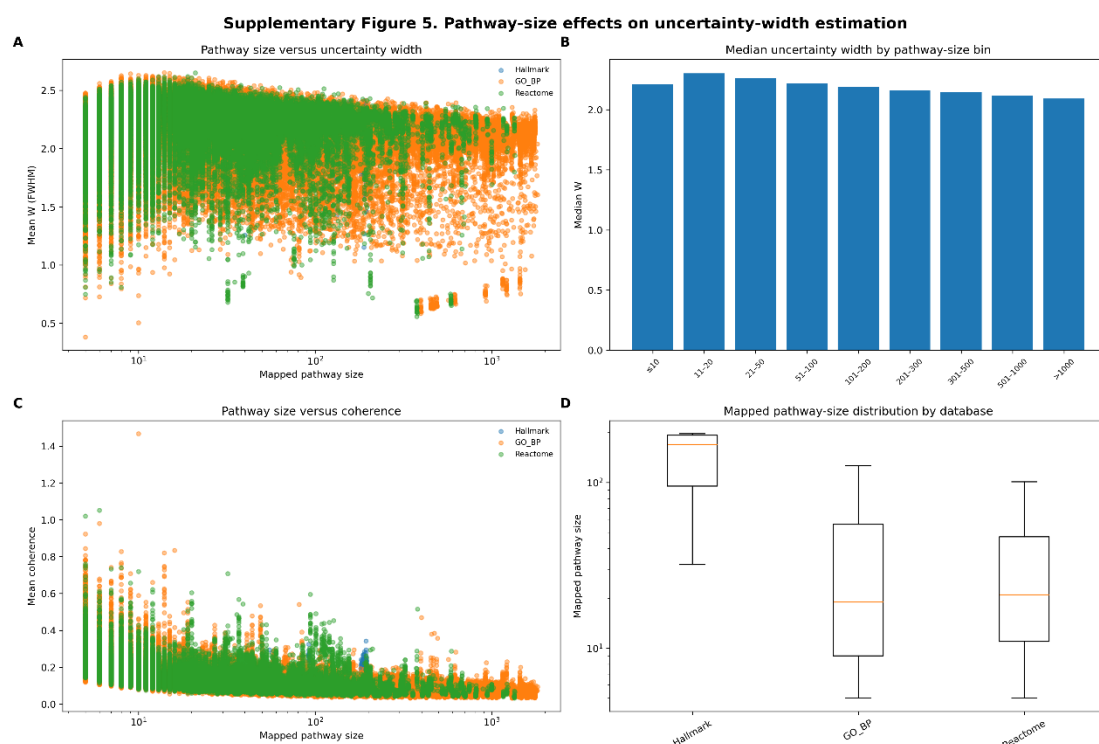
(B) Distribution of optimal observation scales K^* identified across cancers and biological databases. The resulting K^* values span multiple scales, suggesting that different cancer systems preserve clinically relevant structure at different effective observational resolutions.

(C) Maximum discriminability achieved across K-scale optimization. Several cancers exhibit substantial improvement after scale optimization, whereas others show relatively modest gains, indicating heterogeneous sensitivity to observation-scale organization.

(D) Relationship between optimal observation scale K^* and the corresponding maximum discriminability. The absence of a universal monotonic trend suggests

that increasing observational complexity alone does not guarantee improved discriminability. Instead, clinically useful observable organization emerges from a balance between biological coverage and uncertainty accumulation.

Together, these results support the interpretation that discriminability is scale dependent under partial observability and that optimal observation scales may emerge from uncertainty-constrained observable organization rather than maximal observable inclusion.



Supplementary Figure S5 | Relationship between pathway size, uncertainty width, and observable coherence across biological databases and cancers.

(A) Relationship between mapped pathway size and uncertainty width W across all analyzed pathway observables. Larger pathway sets generally exhibit broader observable-width distributions, indicating that increasing observable complexity is associated with expanded uncertainty structure.

(B) Cancer-wise distribution of pathway sizes across Hallmark, GO:BP, and Reactome observables. The analyzed databases span multiple observational scales, enabling evaluation of scale-dependent uncertainty organization.

(C) Relationship between pathway size and observable coherence, defined as $\frac{1}{W}$

$\mu l/(W + \delta)$. Increasing pathway size is frequently associated with reduced coherence, suggesting that larger observable representations may accumulate projection overlap and uncertainty dispersion.

(D) Representative pathway-size distributions and corresponding uncertainty organization across selected cancers. Although pathway size contributes to uncertainty structure, the relationship is nontrivial and does not fully explain discriminability behavior.

Together, these analyses indicate that pathway size influences observable uncertainty organization, but pathway-size effects alone are insufficient to explain the discriminability structure observed under the D-PHY framework.

7. Structured versus random observable geometry

The structured-versus-random framework further clarifies the relationship between uncertainty geometry and biological organization.

Random transcriptomic observables may occasionally exhibit high discriminability because transcriptomic spaces already contain intrinsic structure.

However, curated biological pathways more consistently produce:

- concentrated uncertainty organization,
- reproducible discriminability,
- coherent observable structure.

Thus, biological pathways are interpreted as reshaping transcriptomic observable geometry into more organized uncertainty structures.

This distinction is important.

The D-PHY framework does not interpret biology as replacing randomness entirely.

Instead, biology constrains and reorganizes the observable geometry of an already structured transcriptomic space.

8. Interpretation boundaries

Several interpretation boundaries must be emphasized.

First, observable geometry does not directly reconstruct latent biological states.

Second, uncertainty width does not represent exact entropy.

Third, discriminability does not imply causal biological mechanisms.

Fourth, observable overlap cannot be eliminated completely because transcriptomic projections remain inherently incomplete.

Accordingly, the D-PHY framework should be interpreted as an operational observable-geometry framework rather than a deterministic latent-state inference system.

9. Summary interpretation

The D-PHY framework interprets transcriptomic pathway observables as uncertainty-bearing projected regions within observable space.

Within this framework:

μ

represents observable location,

W

represents uncertainty width,

C

represents uncertainty compression,

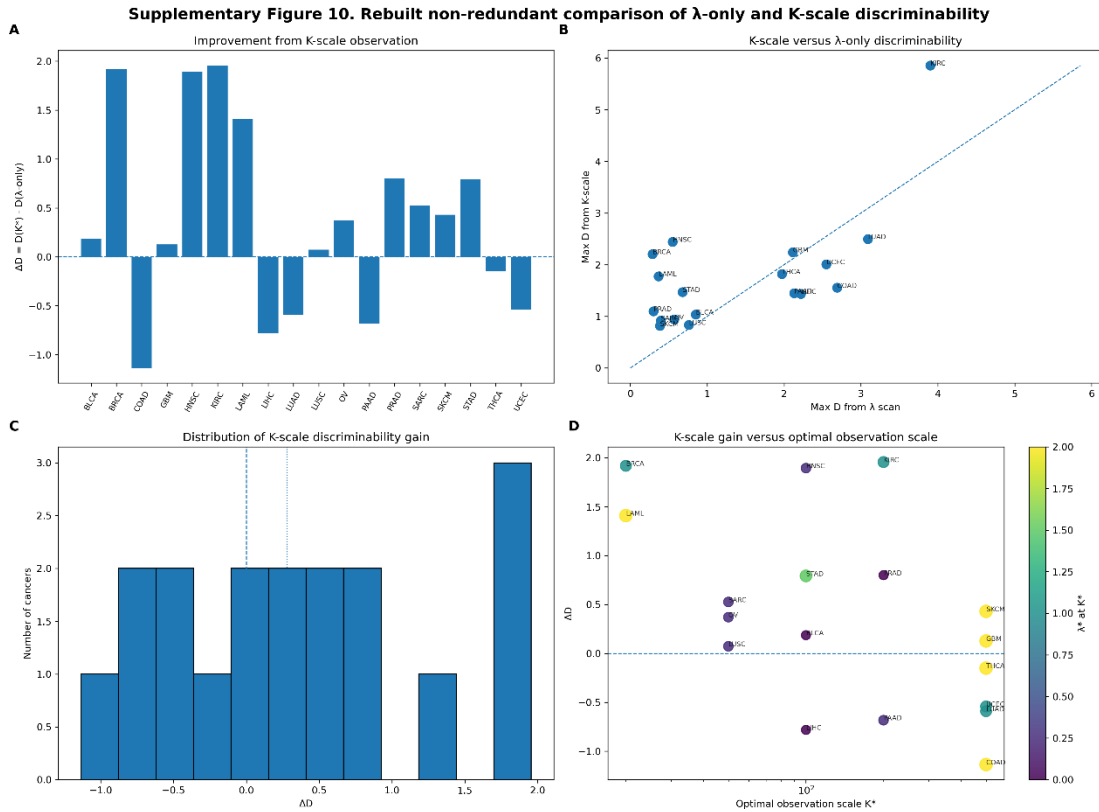
and:

D

represents clinically relevant observable separation.

Under partial observability, clinically useful transcriptomic observables emerge not merely from strong pathway activity, but from sufficiently organized uncertainty geometry that preserves separability despite projection-induced ambiguity.

Thus, discriminability is interpreted as an emergent geometric property of projected observable organization under partial observability.



Supplementary Figure S6 | Integrated comparison between λ -only optimization and K-scale observation optimization across cancers.

(A) Improvement in discriminability obtained after K-scale optimization relative to λ -only optimization. Positive values indicate cancers in which optimizing observation scale K further improves discriminability beyond uncertainty-penalty tuning alone.

(B) Direct comparison between the maximum discriminability obtained from λ -only optimization and that obtained after combined K-scale optimization. Several cancers exhibit improved discriminability under scale-aware observable organization, indicating that observational scale contributes nontrivially to uncertainty-aware discriminability.

(C) Distribution of discriminability gains achieved through K-scale optimization across cancer cohorts. The heterogeneous gain distribution suggests that the importance of observation-scale organization differs substantially across cancer systems.

(D) Relationship between optimal observation scale K^* , optimal uncertainty penalty λ^* , and the resulting discriminability improvement. The observed nontrivial organization indicates that uncertainty penalization and observation scale interact jointly to shape clinically relevant observable structure.

Together, these analyses support the interpretation that uncertainty-aware discriminability depends not only on uncertainty penalization but also on the effective observational scale used to construct pathway-level observable representations.

Supplementary Note S6 | Boundaries and limitations of latent-state inference from transcriptomic projections

One of the central motivations of the D-PHY framework is the recognition that transcriptomic cancer measurements are fundamentally partial observations of highly complex biological systems. This note clarifies the inferential boundaries of the framework and explains why D-PHY should not be interpreted as a direct latent-state reconstruction model.

The central issue is that transcriptomic observables are projections rather than complete representations of the underlying biological system.

Within the D-PHY framework, transcriptomic observations are represented as:

$$O = h(S) + \epsilon$$

where:

- S denotes latent biological organization,
- $h(S)$ denotes the observation mapping,
- O denotes observed transcriptomic measurements,
- ϵ denotes unresolved variability and uncertainty.

Importantly, the observation mapping:

$$h$$

is generally incomplete and non-invertible.

This means that multiple latent biological configurations may generate highly similar transcriptomic observations. Consequently, the inverse problem:

$$O \rightarrow S$$

is intrinsically underdetermined.

This limitation does not arise merely from experimental noise or insufficient data quantity. Instead, it reflects a structural property of projected biological observations.

Accordingly, the D-PHY framework does not attempt to uniquely reconstruct latent tumor states from transcriptomic measurements alone.

This distinction is fundamental.

Many predictive or classification frameworks implicitly assume that sufficiently detailed molecular observations can eventually determine the underlying biological state. However, under partial observability, increasing observational complexity does not necessarily eliminate ambiguity.

Additional observables may simultaneously:

- improve biological coverage,

- increase projection overlap,
- introduce redundancy,
- accumulate uncertainty.

Consequently, more data does not automatically imply more identifiable latent structure.

The D-PHY framework therefore adopts a different objective.

Rather than attempting deterministic latent-state reconstruction, the framework evaluates whether observable organizations preserve clinically meaningful separation under projection constraints.

Within this interpretation, transcriptomic observables are treated operationally as uncertainty-bearing projected regions rather than direct biological-state measurements.

For a biological-process observable B , the framework constructs:

$$(\mu_B, W_B)$$

where:

- μ_B represents observable location,
- W_B represents uncertainty width.

These quantities characterize the geometry of the projected observable rather than the complete latent biological system itself.

Importantly, the framework does not claim that:

$$(\mu_B, W_B)$$

uniquely determines the hidden biological organization.

Instead, the framework evaluates whether certain observable geometries are associated with more stable and discriminable clinical outcome structure.

This is why clinical discriminability:

$$D_{\text{clinical}} = -\log_{10}(p_{\text{log-rank}})$$

is interpreted operationally rather than mechanistically.

High discriminability indicates that an observable organization preserves stronger outcome-associated separation under the current projection framework. It does not imply that the underlying biological system has been uniquely inferred.

This interpretation also explains why uncertainty width:

$$W$$

was introduced.

Under partial observability, two observables with similar average activity may differ substantially in projection ambiguity. A broad observable width may indicate that the transcriptomic projection corresponds to multiple partially overlapping latent biological configurations.

Conversely, narrower observable organization may preserve more concentrated and separable structure.

However, even highly concentrated observable geometry does not guarantee latent-state identifiability.

The D-PHY framework therefore deliberately avoids claims of deterministic state reconstruction.

This distinction is particularly important when interpreting pathway-level transcriptomic observables.

Pathway averages compress:

- gene-level heterogeneity,

- network topology,
- spatial structure,
- regulatory dynamics,
- clonal variation

into reduced effective variables.

Consequently, pathway observables should be interpreted as coarse-grained projections rather than complete mechanistic representations.

Similarly, uncertainty-aware discriminability should not be interpreted as direct biological truth.

The framework evaluates observable usefulness under partial observability, not exact mechanistic causality.

Several practical limitations follow from this interpretation.

First, transcriptomic measurements are static snapshots rather than direct dynamical trajectories. Most public cancer datasets do not contain sufficient longitudinal information to reconstruct:

$$S(t) \rightarrow S(t + 1)$$

explicitly.

Second, bulk transcriptomics compresses cellular and spatial heterogeneity into averaged measurements. Distinct spatial organizations may therefore produce similar transcriptomic projections.

Third, pathway observables ignore many forms of biological structure, including:

- signaling directionality,
- protein interaction topology,
- regulatory hierarchy,
- metabolic coupling,

- temporal adaptation.

Fourth, survival-based evaluation itself is indirect. Survival reflects many downstream influences beyond transcriptomic organization alone.

For these reasons, the D-PHY framework should be interpreted conservatively.

The framework does not claim:

- exact tumor-state reconstruction,
- deterministic progression inference,
- complete biological observability.

Instead, the framework provides an operational strategy for studying how uncertainty-bearing transcriptomic observables preserve clinically relevant structure under projection constraints.

Within this interpretation, the central object of study becomes observable geometry rather than exact latent-state inference.

This distinction also explains why the framework emphasizes:

- uncertainty width,
- coherence,
- discriminability,
- scale dependence,
- structured-versus-random organization.

These quantities characterize observable organization under partial observability without requiring complete inversion of the latent biological system.

Importantly, the present limitations do not imply that transcriptomic analysis is biologically meaningless. Rather, they imply that transcriptomic observables should be interpreted probabilistically and geometrically rather than

deterministically.

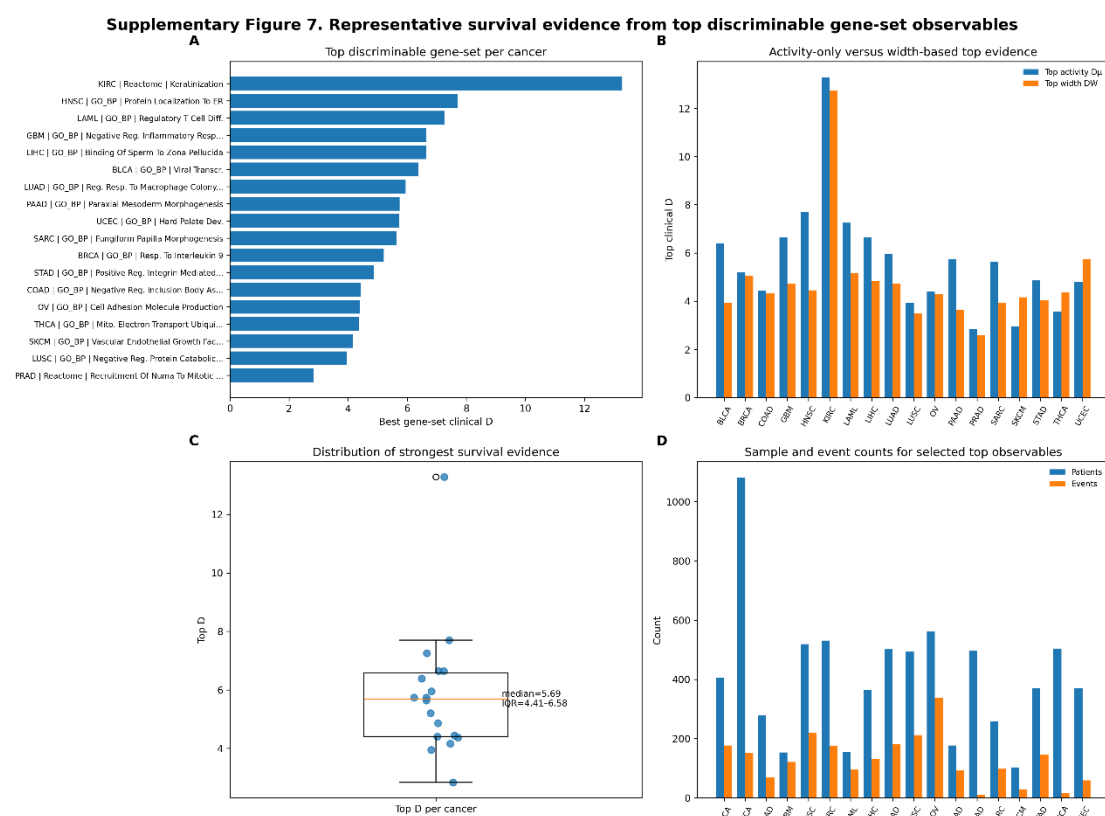
Future developments may partially reduce these limitations through incorporation of:

- longitudinal measurements,
- spatial transcriptomics,
- single-cell sequencing,
- temporal treatment trajectories,
- network-constrained observables,
- multimodal biological integration.

Such approaches may improve observable fidelity and reduce projection ambiguity.

Nevertheless, even under richer observational settings, complete latent-state reconstruction may remain intrinsically limited because biological observation itself is fundamentally projected and partially observable.

Within this broader context, the D-PHY framework should therefore be interpreted as an uncertainty-aware observable-geometry framework designed to evaluate clinically relevant transcriptomic organization under partial observability, rather than as a deterministic latent-state reconstruction system.



Supplementary Figure S7 | Representative survival evidence from top discriminable gene-set observables across cancers.

(A) Top discriminable pathway-level observables identified across cancer cohorts under the D-PHY framework. Each entry represents the highest-performing observable within a given cancer cohort, together with its associated biological database and pathway identity.

(B) Comparison between activity-based discriminability (D_μ) and width-aware discriminability (D_W) for the strongest observable identified in each cancer cohort. Several cancers exhibit improved discriminability when uncertainty width W is incorporated, indicating that uncertainty-aware observable representations capture clinically relevant structure beyond activity magnitude alone.

(C) Distribution of the strongest discriminability values obtained across cancers. The resulting distribution demonstrates substantial heterogeneity in achievable observable separation under partial observability, suggesting that clinically relevant observable organization differs across cancer systems.

(D) Patient and event counts associated with the representative top-performing observables. These analyses provide additional context regarding cohort size and

survival-event availability underlying the reported discriminability behavior. Together, these analyses provide representative clinical evidence supporting the operational relevance of uncertainty-aware pathway-level observables under the D-PHY framework.

Supplementary Note S7 | Future directions and implications of uncertainty-aware observable analysis under partial observability

The D-PHY framework was developed to analyze transcriptomic observables under conditions of partial observability. Rather than attempting deterministic latent-state reconstruction, the framework evaluates how observable organization, uncertainty structure, and discriminability emerge from projected biological measurements.

This note discusses the broader implications of the framework and outlines several future directions for uncertainty-aware observable analysis in complex biological systems.

Importantly, the purpose of this section is not to claim that the present framework solves latent-state inference in cancer systems. Instead, the objective is to clarify why uncertainty-aware observable analysis remains valuable even when complete biological identifiability is unattainable.

1. Why transcriptomic observations alone are insufficient

Bulk transcriptomic measurements provide only partial access to biological organization.

A transcriptomic profile reflects the combined contribution of:

- tumor-cell populations;
- stromal components;

- immune infiltration;
- clonal heterogeneity;
- unresolved spatial organization;
- regulatory interactions;
- microenvironmental dynamics.

Consequently, transcriptomic observations compress complex biological structure into projected measurements.

Within the D-PHY framework, this is represented as:

$$O = h(S) + \epsilon$$

where the latent biological organization S remains only partially observable.

Under such conditions, transcriptomics alone is unlikely to uniquely reconstruct the complete biological state of a tumor.

This limitation is structural rather than merely technical.

Increasing observational complexity may improve biological coverage, but it may also introduce:

- projection redundancy;
- observable overlap;
- uncertainty accumulation;
- dimensional compression.

Accordingly, the central challenge is not simply obtaining more data, but understanding how biologically relevant structure survives under projection constraints.

This is one of the primary motivations for uncertainty-aware observable analysis.

2. Why uncertainty-aware observables remain useful

The inability to reconstruct complete latent biological states does not imply that transcriptomic observables are biologically meaningless.

In many areas of science and medicine, useful operational observables exist despite incomplete system observability.

Examples include:

- effective thermodynamic variables;
- imaging-based diagnostic markers;
- clinical laboratory measurements;
- phenomenological physical models.

Such observables remain valuable because they preserve useful structure even without complete mechanistic reconstruction.

The D-PHY framework adopts a similar perspective.

The goal is not to eliminate uncertainty entirely, but rather to evaluate:

- which observable organizations remain informative,
- how uncertainty structure influences discriminability,
- how observable geometry affects clinical separability.

Within this interpretation, transcriptomic pathway observables become operational tools for studying projected biological organization under uncertainty.

3. Toward longitudinal uncertainty-aware trajectory analysis

The current D-PHY framework primarily analyzes static transcriptomic observations linked to downstream clinical outcomes.

However, the long-term motivation extends naturally toward trajectory-based

analysis.

Ideally, future frameworks would analyze explicit biological evolution:

$$S(t) \rightarrow S(t + 1) \rightarrow S(t + 2)$$

rather than relying primarily on static survival-associated separation.

Such trajectory-oriented analysis may become possible through:

- longitudinal transcriptomics;
- treatment time-course datasets;
- serial biopsies;
- adaptive therapy measurements;
- temporal single-cell analysis.

Under such settings, uncertainty-aware frameworks may eventually evaluate:

- trajectory divergence;
- uncertainty evolution;
- attractor transitions;
- treatment-induced observable reorganization.

The present D-PHY framework should therefore be viewed as an initial observable-geometry foundation rather than a final dynamical-state model.

4. Potential integration with spatial and single-cell observations

One major limitation of bulk transcriptomics is the compression of spatial heterogeneity into averaged measurements.

Future uncertainty-aware frameworks may benefit substantially from:

- spatial transcriptomics;
- spatial proteomics;
- single-cell sequencing;
- spatial imaging integration.

Such measurements may reduce projection ambiguity by preserving local organizational structure.

For example, spatially resolved observables may allow uncertainty geometry to be studied not only in transcriptomic space, but also across:

$$(x, y, z, t)$$

biological organization.

Similarly, single-cell measurements may reveal uncertainty structure that is hidden within bulk averages.

However, even under richer observational technologies, complete biological identifiability may remain intrinsically limited because biological observation itself remains projected and scale dependent.

5. Relationship to complex-systems analysis

The D-PHY framework shares several conceptual connections with complex-systems analysis.

Complex biological systems commonly exhibit:

- emergence;
- heterogeneity;
- multi-scale organization;
- coarse-grained observability;
- uncertainty accumulation;
- partial observability.

Under such conditions, effective observables often become more practical than exact microscopic reconstruction.

The D-PHY framework therefore emphasizes:

- observable organization,
- uncertainty geometry,
- scale dependence,
- operational discriminability

rather than exact deterministic inversion of latent biological states.

Importantly, the framework does not claim that cancer systems can be fully reduced to simple statistical descriptions. Rather, the framework studies how projected observables behave under incomplete access to latent organization.

6. Potential future observable layers

The present study focused primarily on transcriptomic pathway observables derived from gene-expression data.

However, the uncertainty-aware observable framework may potentially be extended to additional observable layers, including:

- copy-number variation;
- proteomics;
- phosphoproteomics;
- imaging-derived observables;
- pathology features;
- spatial molecular measurements;
- multimodal integrated observables.

Different observable modalities may preserve different aspects of latent biological organization.

Future work may therefore investigate how uncertainty geometry changes across different observable spaces.

7. Structured observable construction beyond pathway averaging

The current framework uses pathway-level averaging as a biologically interpretable coarse-graining strategy.

Future extensions may incorporate more structured observable construction, including:

- network topology;
- pathway connectivity;
- protein–protein interaction structure;
- signaling directionality;
- graph-based pathway observables.

Such approaches may partially reduce projection ambiguity by incorporating biological structural constraints directly into observable construction.

However, even highly structured observables will likely remain partially projected representations rather than complete latent-state reconstructions.

8. Why D-PHY remains useful despite inferential limitations

An important conceptual point is that uncertainty-aware observable analysis remains useful even if complete latent-state reconstruction is impossible.

Clinical medicine itself frequently operates using incomplete but useful observables.

Examples include:

- radiographic interpretation;
- laboratory biomarkers;
- physiological measurements;
- symptom-based diagnosis.

Such observables remain clinically valuable because they preserve actionable

structure despite incomplete mechanistic access.

Similarly, the D-PHY framework evaluates whether transcriptomic observables preserve:

- clinically relevant separation,
- coherent uncertainty organization,
- reproducible observable structure.

Under this interpretation, observable usefulness does not require perfect biological identifiability.

Instead, the objective becomes understanding how informative observable organization survives under uncertainty-bearing projection.

9. Broader implications of uncertainty-aware observable analysis

The broader implication of the D-PHY framework is that biological analysis may benefit from explicitly incorporating uncertainty geometry into observable interpretation.

Traditional transcriptomic analysis often emphasizes:

- activity magnitude,
- differential expression,
- clustering,
- statistical significance.

The D-PHY framework suggests that observable width, overlap, and uncertainty structure may also play important roles in determining observable usefulness.

This perspective shifts the analytical focus from:

“Which observable changes?”

toward:

“Which observable organization remains separable under projection uncertainty?”

Such a shift may become increasingly important as biological datasets grow in complexity and dimensionality.

10. Concluding perspective

The D-PHY framework does not attempt to eliminate uncertainty from transcriptomic analysis.

Instead, the framework studies how clinically relevant structure survives within uncertainty-bearing projected biological observables.

Within this interpretation:

- pathway observables are projected effective variables;
- uncertainty width represents observable spread;
- coherence represents uncertainty compression;
- discriminability represents operational separability under projection constraints.

The central goal is therefore not exact latent-state reconstruction, but rather the characterization of observable organization under partial observability.

From this perspective, uncertainty is not treated merely as noise to be removed, but as an intrinsic structural component of projected biological observation itself.